14 Write $0.6\dot{2}\dot{4}$ as a fraction in its simplest form.

Let $x = 0.6242424\ldots$ (the block 24 repeats).

$$1000x = 624.242424\ldots$$

$$10x = 6.242424\ldots$$

1 mark
Align repeating tails

Subtract: $990x = 618$

1 mark
Subtract to eliminate recurring decimals

$$x = \frac{618}{990} = \frac{103}{165} \quad \text{(divide by 6)}$$

1 mark
103/165

(Total for Question 14 is 3 marks)